Course Syllabus

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Course Description

Mathematics provides the foundation for reasoning, problem-solving, and analysis in computer science. Geometry and Calculus for Computing equips you with essential tools to model shapes, describe motion, and analyse change. Across four modules, you'll build a solid grounding in trigonometry, graph sketching, kinematics, exponential and logarithmic functions, and introductory calculus. You'll learn to connect abstract mathematical concepts to practical computing applications, from computer graphics and simulations to optimisation and algorithm analysis. By the end of the course, you'll have the skills to interpret functions, calculate gradients, and apply mathematical reasoning to a wide range of computational problems. As the third course in the Essential Mathematics for Computer Science specialisation, this course builds upon your understanding of discrete structures and advanced methods, extending your mathematical toolkit to include continuous mathematics essential for graphics, physics simulations, and algorithm optimization.

Target Audience

This course is designed for computer science students, software developers, data scientists, and professionals working in computational fields who need to understand and apply geometric and calculus concepts. It's ideal for:

- Computer science and engineering students seeking to strengthen their mathematical foundation
- Software developers working in graphics, simulation, or scientific computing
- Data scientists and analysts who need to model continuous processes and changes
- AI and machine learning practitioners requiring calculus for optimization algorithms
- Anyone seeking to understand the mathematical principles behind computational visualization and modeling

A basic understanding of algebra and coordinate systems is recommended, with prior exposure to functions and mathematical notation (such as that covered in earlier specialization courses) being beneficial but not strictly required.

Content Differentiation

This course distinctively bridges the gap between abstract mathematical theory and practical computational applications. Unlike traditional mathematics courses, it focuses specifically on the geometric and calculus concepts most relevant to computing, with:

- Visual and interactive approaches to trigonometry and calculus that build intuition
- Direct connections between mathematical concepts and their computing applications
- Emphasis on computational tools and representations for mathematical concepts
- Progressive development from basic principles to applied problem-solving
- Integration of digital tools like Desmos for exploration and visualization

Each concept is introduced with clear computing relevance—from trigonometric functions in computer graphics and animation to calculus in optimization algorithms and physics simulations—ensuring you understand not just how to solve problems mathematically, but why these techniques matter in computing contexts.

Course Goals and Objectives

Upon successful completion of this course, you will be able to:

- Apply trigonometric principles to solve problems in computational geometry and graphics
- Sketch and interpret function graphs to model computational processes and data relationships
- Use kinematics to describe and predict motion in simulations and animation systems
- Work with exponential and logarithmic functions in computational growth models and complexity analysis
- Calculate derivatives to determine rates of change and optimize functions
- Apply limits to understand function behavior and continuity in computational contexts
- Implement mathematical models using appropriate computational tools and techniques
- Translate between mathematical representations and computing implementations

Course Outline

The course consists of four modules that progressively build your understanding of geometry and calculus for computing:

Module 1: Angles, Triangles and Trigonometry Introduces fundamental geometric concepts, trigonometric ratios, and their applications in computing, providing essential tools for spatial reasoning and graphics.

Module 2: Graph Sketching and Kinematics Explores function visualization, Cartesian coordinates, and the mathematical description of motion, connecting abstract functions to practical computing applications.

Module 3: Exponential and Logarithmic Functions Examines special functions with unique properties that model growth, decay, and scaling relationships crucial in numerous computing applications.

Module 4: Limits and Differentiation Introduces foundational calculus concepts, focusing on how to quantify change and optimize functions—essential skills for algorithm analysis and computational modeling.

Learning Objectives

Module 1: Angles, Triangles and Trigonometry

By the end of this module, you will:

- Represent and work in the 2-dimensional plane
- Correctly identify and label lines, angles, triangles, and circles
- Apply the basics of trigonometry—sine, cosine, and tangent
- Understand and use relationships among elements of triangles and circles
- Apply trigonometric relations to computational geometry problems
- Convert between degrees and radians for computational applications
- Use Pythagoras' theorem and trigonometric ratios in combination

Module 2: Graph Sketching and Kinematics

By the end of this module, you will:

- Define functions and understand their representation in Cartesian coordinates
- Identify and work with inverse functions
- Plot linear and polynomial functions on the Cartesian plane
- Analyze higher-order functions and understand their limits
- Apply transformations to functions and interpret their graphical meaning
- Use kinematics to describe uniform and uniformly accelerated motion
- Implement mathematical models of motion in computational simulations

Module 3: Exponential and Logarithmic Functions

By the end of this module, you will:

- Define exponential functions and derive their basic properties
- Define logarithms as inverse functions of exponentials and derive their properties
- Plot and interpret exponential and logarithmic functions
- Solve equations involving exponentials and logarithms
- Apply these functions to computational growth models and scaling problems
- Recognize and utilize these functions in algorithm complexity analysis

Module 4: Limits and Differentiation

By the end of this module, you will:

- Determine limits of sequences and functions
- Use limits to identify asymptotic behavior of functions
- Distinguish between continuous and discontinuous functions
- Calculate derivatives of functions from first principles
- Apply product, quotient, and chain rules for differentiation
- Use limits and derivatives to find local minima, maxima, and turning points
- Apply differentiation techniques to optimization problems in computing

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